

Example of Questions

- Points pre-assigned to review questions and problems are just an indication of the final score: partial and unsatisfactory answer/solution will receive a fraction of the points, whereas excellent answers may exceed the indicated points.
- Use of books/notes is forbidden.
- Indicate name and UC ID number in ALL the solution sheets. Number the solution sheets and indicate total number of sheets (*e.g.*, first sheet 1/3, second sheet 2/3, third sheet 3/3).

Questions will include theory material and proofs, but not full problems.

R1 (5Points) Consider two random variables X_1 and X_2 , distributed according to an Exponential distribution with parameters λ_1 and λ_2 . The expectation of the random variable $X = \min(X_1, X_2)$ is

- $\lambda_1 + \lambda_2$;
- $\min(\lambda_1, \lambda_2)$;
- $1/(\lambda_1 + \lambda_2)$;
- $1/\min(\lambda_1, \lambda_2)$;

R2 (XPoints) Describe multiplexing and demultiplexing of TCP and UDP protocols.

R3 (XPoints) Describe the slotted Aloha protocol. Does the protocol provide adaptation to congestion? Explain your answer.

R4 (XPoints) Discuss why hierarchical routing is used in the internet (instead of flat 1-level routing).

R4 (XPoints) Describe the operations of the slow start and congestion avoidance states in TCP.